

Task 1 - Brief Explanation of the Concept Map

The concept map presents the fundamental concepts of cloud computing and explains how they are connected to the SentinelX Cloud Threat Intelligence Using NLP and Serverless Risk Scoring project. Cloud Fundamentals form the base of the system by providing on-demand services, scalability, elasticity, and measured service. Cloud Evolution highlights the technological developments that made cloud platforms practical for modern applications. The Architecture and Deployment section illustrates the interaction between the frontend, backend, network, and cloud infrastructure. The Service Models (IaaS, PaaS, SaaS, and FaaS/Serverless) define the responsibilities shared between the cloud provider and the development team. Virtualization enables efficient resource sharing through virtual machines, containers, and resource pooling. Web Services and REST APIs allow secure communication between the web application, NLP engine, serverless risk-scoring functions, and the cloud database. Finally, the capstone connection demonstrates how SentinelX uses NLP to analyze cyber threat data, serverless computing to calculate risk scores, and cloud storage and dashboards to provide secure, scalable, and real-time threat intelligence.